



Improving Evidence Retrieval for Scientific QA via Rich Contextualization and Domain Adaptation

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CS 5624, Spring

2026

RESEARCH QUESTION

Central problem: Automated evidence retrieval in document-level scientific QA. Given a peer-review question and a paper averaging 11,700 tokens, identify the exact paragraph or sentence containing the answer.

Why retrieval matters: In a RAG pipeline, retrieval recall correlates with generation quality ($r = 0.42$). Improving retrieval is the highest-leverage intervention.

Three gaps we target:

- **Lexical gap** — Reviewers use informal shorthand; papers use precise technical terms; BM25 needs exact-term overlap
- **Zero-shot limitation** — Dense retrievers never fine-tuned on scientific text
- **Token overwhelming** — Prepending only the paper title to short sentences drowns the evidence signal; sentence MRR = 0.36 vs. paragraph MRR = 0.48

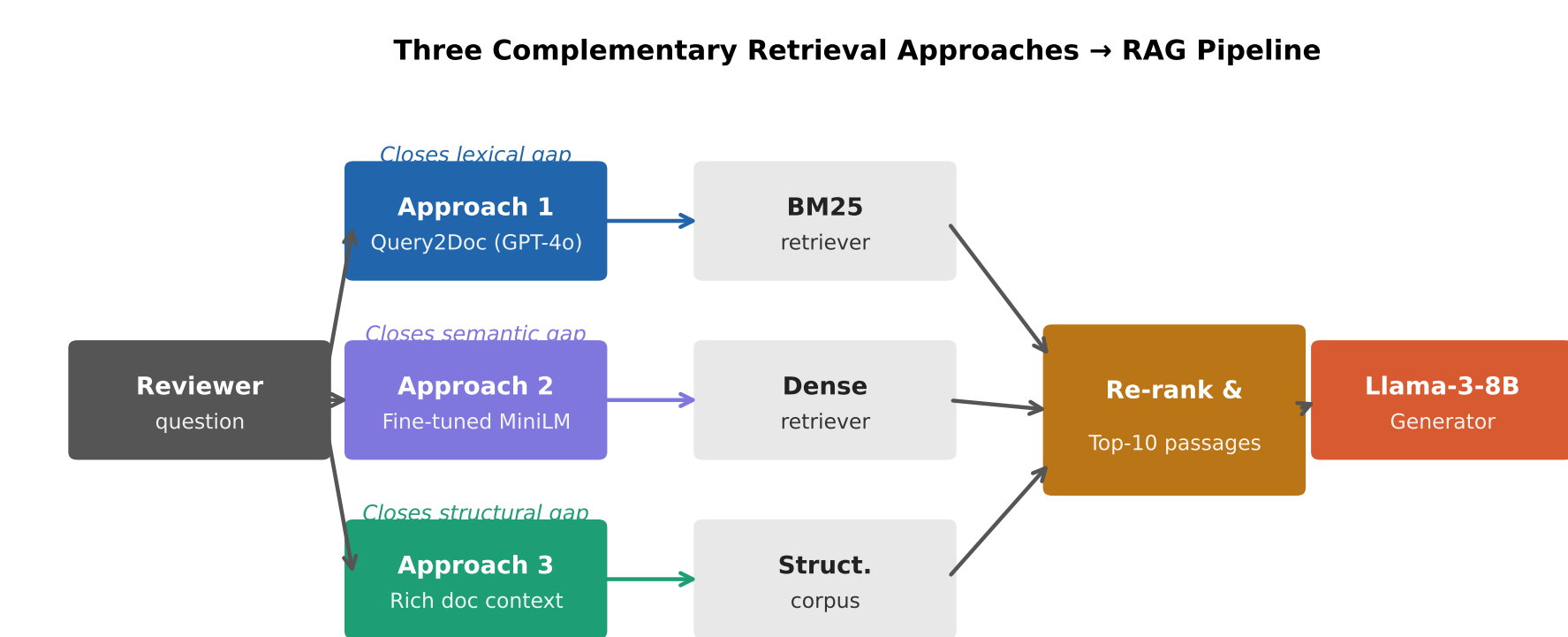
DATASET: PEERQA

PeerQA (NAACL 2025 Outstanding Paper) — natural questions from peer reviews, evidence annotated by paper authors — the highest possible annotation authority.

Annotated QA pairs	579
Academic papers	208
Avg. paper length	~11,700 tok.
Answerable questions	383
Non-consec. evidence	~30%
Unlabeled questions	11,967
Domains	ML, NLP, Geoscience

SYSTEM PIPELINE

All three approaches are **complementary** — each targets a different failure mode.



EVALUATION METRICS

MRR — Mean Reciprocal Rank of first correct passage.

Recall@K — correct passage in top K (critical for RAG).

Setup: 383 answerable questions; retrieval is *per paper* (rank within the question's source document only), ~28.5k paragraph chunks over 167 papers — matches PeerQA's published protocol.

Published baselines: Dragon⁺ MRR = 0.4845; BM25 MRR = 0.4288; MiniLM = 0.3654.

ACKNOWLEDGMENTS & REFERENCES

[1] Baumgartner et al., *PeerQA*, NAACL 2025. We acknowledge the UKP Lab at TU Darmstadt for the PeerQA dataset.

APPROACH 1: BM25 + QUERY2DOC

Problem: BM25 requires exact-term overlap, but reviewer questions are phrased informally.

Key insight: instead of paraphrasing the question, prompt GPT-4o to *generate a hypothetical answer* in the formal academic language of the paper. Augmented query = (question×5) + pseudo-doc. Total API cost: \$1.10 for 383 questions.

Why not paraphrasing? It keeps question register, dropping technical terms. Query2Doc adds discriminative vocabulary (“convergence rates,” “ablation study”) that directly increases BM25 overlap with the evidence.

Method	MRR	R@1	R@5	R@10
BM25 (orig.)	0.3852	0.1929	0.4785	0.6072
BM25 + RRF	0.4174	0.2278	0.4913	0.6323
BM25 + Q2D	0.4432	0.2500	0.5086	0.6527
PeerQA BM25 (pub.)	0.4288	—	—	0.6817

Result: surpasses published BM25 baseline by **+1.44 MRR points**; R@1 improves **+29.6%**. RRF hurts: when one ranker dominates, fusion pulls performance toward the weaker component.

APPROACH 2: FINE-TUNED MINILM

Problem: dense retrievers trained on web text; never exposed to scientific question-evidence structure.

Method: fine-tune all-MiniLM-L6-v2 on 383 labeled pairs using MultipleNegativesRankingLoss with **same-paper hard negatives**. 3-fold cross-validation, 1 epoch per fold.

Why same-paper negatives? Random negatives are trivially easy. Same-paper negatives force the model to separate method vs. results vs. background — the actual difficulty of PeerQA.

Method	MRR	R@1	R@5	R@10
MiniLM (zero-shot)	0.3286	0.1570	0.3766	0.5360
MiniLM (sup., 3-fold)	0.3618	0.1722	0.4207	0.5754
Contriever (zs, pub.)	0.3494	—	—	—
Dragon ⁺ (pub.)	0.4845	—	—	0.6817

Result: fine-tuned MiniLM **outperforms the larger zero-shot Contriever**. Largest gain at R@5 (+11.7%), the window most useful for RAG. 3-fold MRR standard deviation ± 0.0196 — moderate spread, as expected with a small labeled set.

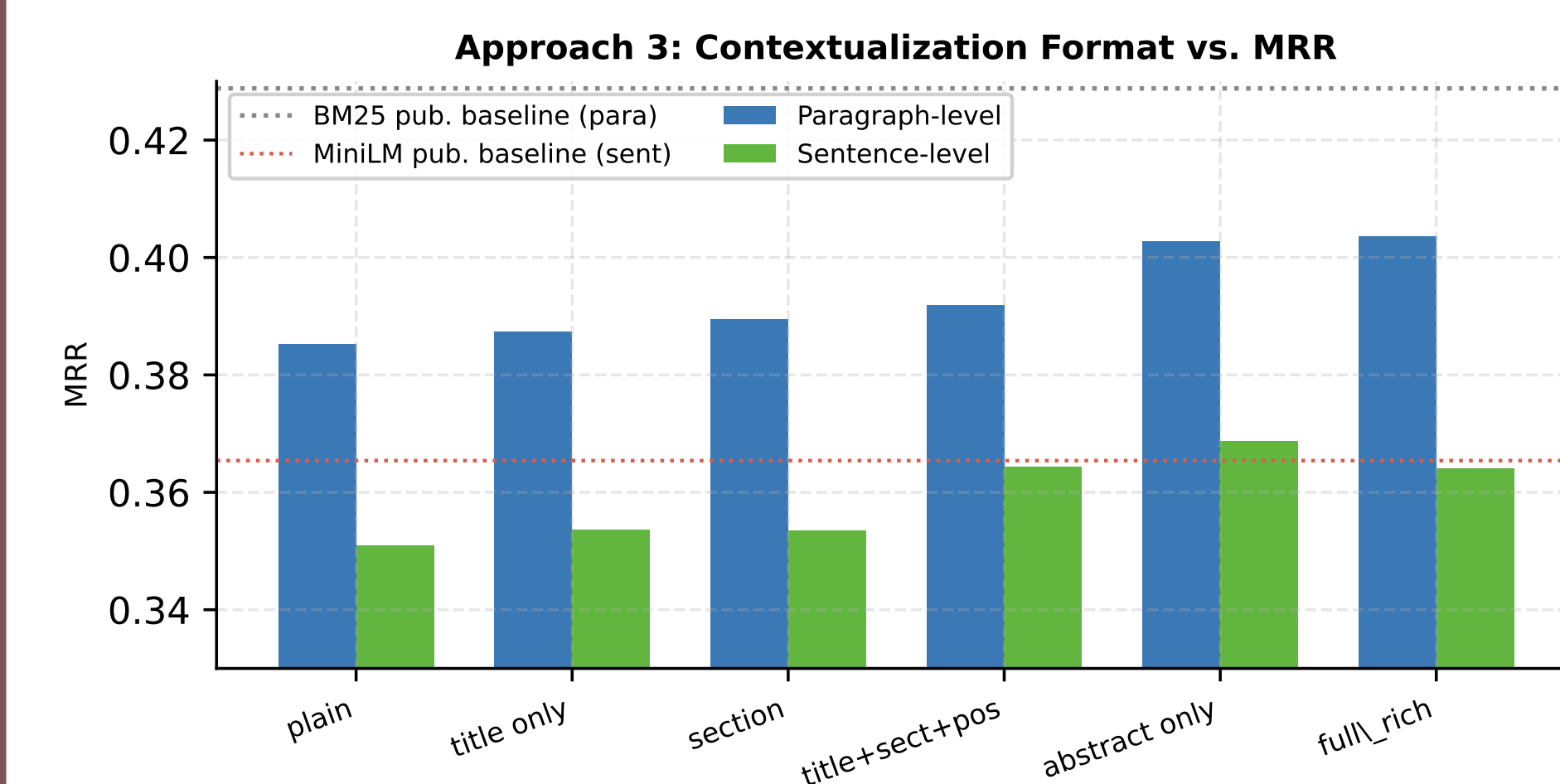
APPROACH 3: STRUCTURAL CONTEXT

Problem: prepending only the title overwhelms short sentences.

Granularity matters: *full_rich* wins paragraph MRR; *abstract_only* wins sentence MRR — richer prefixes help long units but re-introduce token overwhelming on short sentences.

Method: test 7 context-string templates prepended to each retrieval unit. Zero API cost; ~50 seconds CPU runtime.

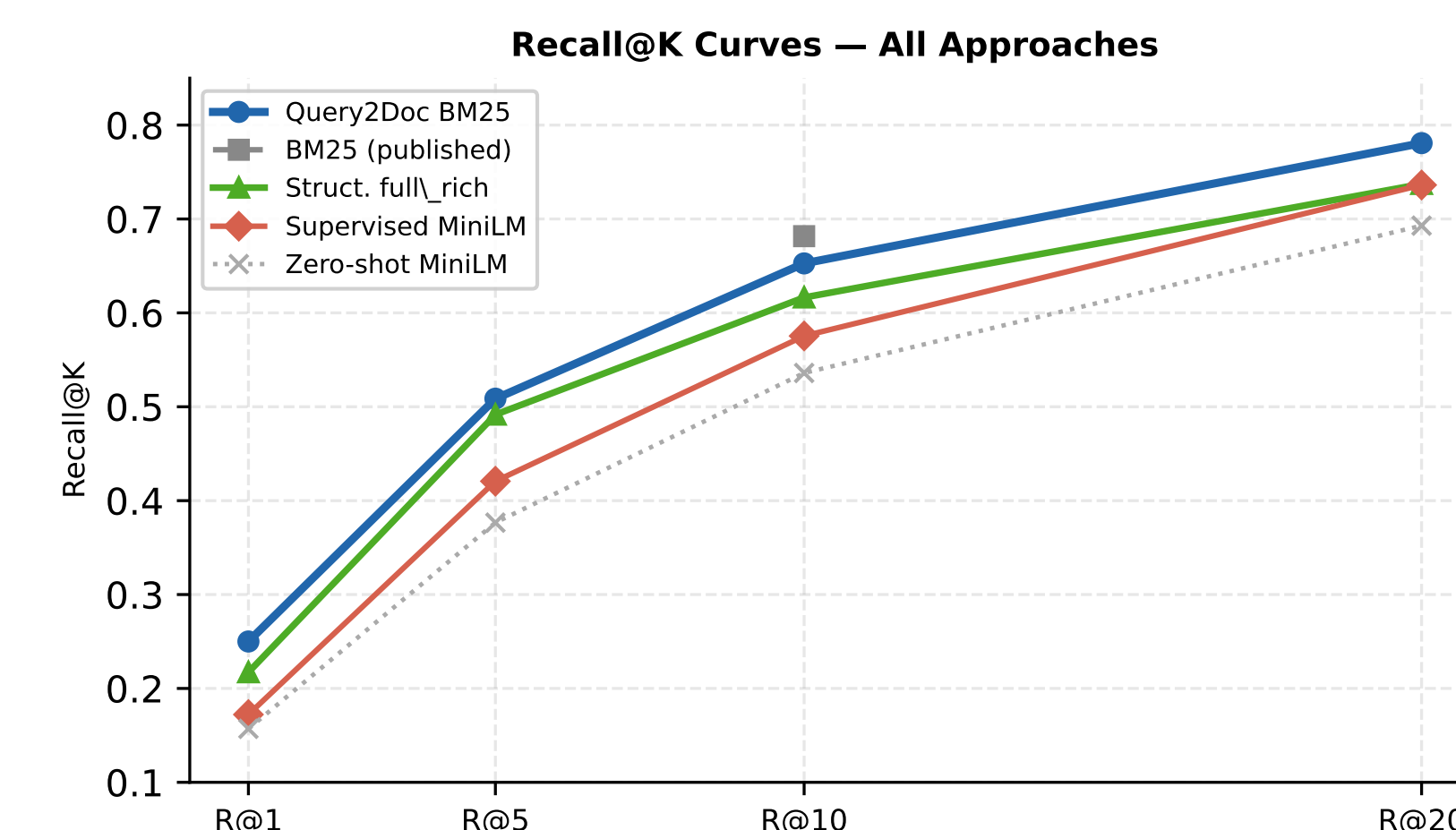
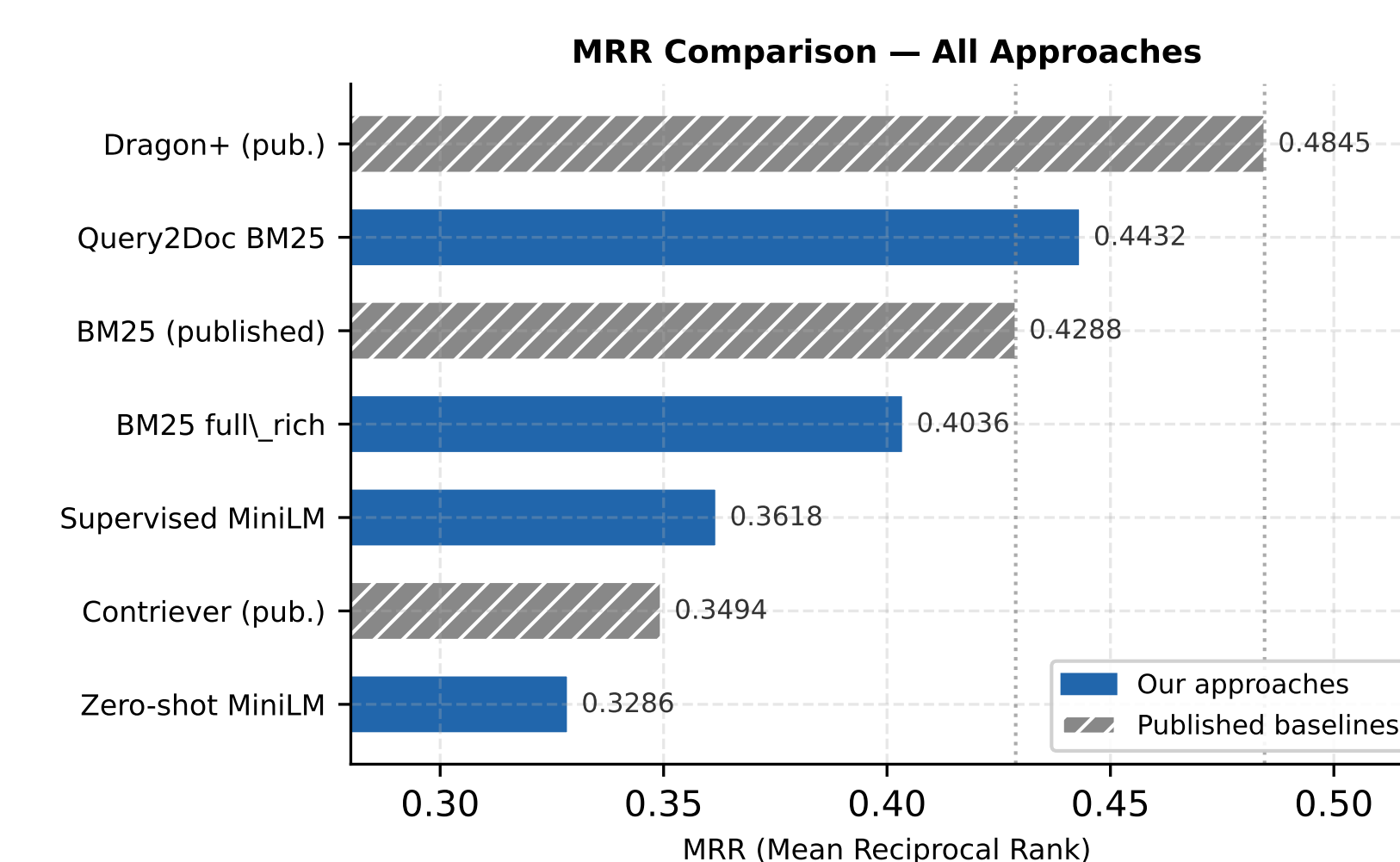
[abstract sent.] | [Title] |
[Section] | [Para X, Sent Y] | [text]



Key finding: the abstract's first sentence is the single most valuable component. At *sentence level*, adding too many components re-introduces token overwhelming — use *abstract_only*. At *paragraph level*, *full_rich* is optimal.

Format	Para MRR	Sent MRR
plain	0.3852	0.3509
title only	0.3873	0.3536
abstract only	0.4027	0.3687
full_rich	0.4036	0.3640
PeerQA pub.	0.4288	0.3654

RESULTS: MRR & RECALL



CONCLUSIONS & FUTURE WORK

- **Q2D** strongest here (MRR 0.4432; +0.0144 vs pub. BM25). Hypothetical answer text beats paraphrase for BM25 overlap.
- **Context** MRR 0.4036, \$0 API; pick template by para. vs sentence granularity.
- **MiniLM sup.** 0.3618 > Contriever zs 0.3494; same-paper negatives match PeerQA difficulty.
- **Next:** combine Q2D + *full_rich* + dense re-rank; Llama-3-8B RAG; ROUGE-L / AlignScore.